

DoView Tool J6 - Mathematization of Outcomes Theory and DoView Planning

Any theory, such as outcomes theory and its applied version DoView Planning can benefit from exploring how mathematization could help formalize it and provide useful tools when using it in practice. Here are some possible examples.

- Category theory: Could provide a robust way of thinking about DoView boxes and the relationships between them.
- Graph/network theory: Views DoView diagrams as nodes and links, including connectivity, cycles, and influence.
- Matrix and linear algebra: Could represent the strength of links using matrices and help calculate which outcomes are most influential. Eigen-analysis could potentially show key pathways or bottlenecks.
- Probabilistic graphical models: Bayesian networks could be used to assign probabilities to links, estimating the likelihood of different outcomes and supporting forecasting under uncertainty.
- Markov chains/state-transition matrices: Could be useful when each step depends on the previous one, such as staged interventions. Shows how likely the system is to move from one outcome to another over time.
- System dynamics: Uses stocks, flows, and equations to represent how steps and outcomes build up or are depleted, including feedback loops and time delays.
- Causal inference/do-calculus: Tests if the causal links in a DoView diagram are supported by real data and could help identify hidden factors.
- Multi-criteria decision analysis/utility theory: Assigns scores to different outcomes, could help to weigh options and make choices when trade-offs are needed.
- Logic formalisms (temporal/description logic): Enables rule-based and time-based reasoning in DoView diagrams and could check for logical consistency and sequencing.
- Multiple regression: Estimates the influence of several factors on an outcome in a DoView diagram, showing how different variables together predict results.
- Factor analysis: Could potentially identify underlying patterns or groups among many boxes in a DoView diagram, helping to simplify complex relationships.